

Unit Outline

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COS30018

Intelligent Systems

Semester May 2026

Please read this Unit Outline carefully. It includes:

PART A Unit summary

PART B Your Unit in more detail

PART C Further information



PART A: Unit Summary

Unit Code(s)		COS30018							
Unit Title		Intelligent Systems							
Duration		One semester or equivalent							
Total Contact Hours		48 hours							
Requisites:									
	Pre-requisites	COS20011 Software Development in Java OR COS20007 Object-Oriented Programming OR COS30016 Programming in Java OR 65+% in SWE20004 Technical Software Development							
	Co-requisites	Nil							
	Concurrent pre-requisites	Nil							
	Anti-requisites	Nil							
	Assumed knowledge	Object-oriented programming at intermediate level							
Credit Points		12.5							
Campus/Location		Hanoi - HCM - Danang							
Mode of Delivery		Online and Face-to-Face							
Assessment Summary		<table><tr><td>1. Assignment</td><td>Group</td><td>50%</td></tr><tr><td>2. Continuous Oral Defence</td><td>Individual</td><td>50%</td></tr></table> As the minimum requirements of assessment to pass a unit and meet all ULOs to a minimum standard, a student must achieve an aggregate mark of 50% or more.		1. Assignment	Group	50%	2. Continuous Oral Defence	Individual	50%
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2. Continuous Oral Defence	Individual	50%							

Aims

This Unit is designed to introduce students to a range of artificial intelligence (AI) techniques and how to apply them to solve real-world problems.

Unit Learning Outcomes

Students who successfully complete this unit can:

1. Explain a range of techniques of intelligent systems across artificial intelligence (AI) and intelligent agents (IA); both from a theoretical and a practical perspective.
2. Apply different AI/IA algorithms to solve practical problems.
3. Design and build simple intelligent systems based on AI/IA concepts.

Graduate Attributes

The Swinburne Graduate Attributes describe the capability of our graduates to use knowledge, skills and behaviours to contribute to society meaningfully and positively. They include professional, self-directed learning and future-ready skills.

This unit contributes to the development of the following Swinburne Graduate Attributes:

- *GA1 Communication - Verbal communication:* Students are required to attend multiple oral defence sessions to demonstrate their understanding of the subject matter.
- *GA2 Communication - Communicating using different media:* Students are required to write reports (for the assignments) to discuss their understanding of the problems as well as their approaches for solving the problems. They are also required to provide comments to explain their code.
- *GA3 Teamwork - Collaboration and negotiation:* Students are required to work within a team for Assignment 2.
- *GA5 Digital literacies– Information literacy:* Students are required to perform research on the Internet to learn about algorithms and how to implement them to fulfil the requirements of the Assignments.
- *GA6 Digital Literacies– Technical literacy:* Students are required to use a programming language (e.g., Python) to implement algorithms learned from this unit and deliver a working software.

Other graduate attributes may be practised in the unit but are not formally taught as part of the unit content, nor incorporated within formal assessment.

- *GA4 Teamwork – Teamwork roles and processes.*

Content

The unit includes the following topics:

- Introduction to Intelligent Systems
- Knowledge representation and reasoning
- Intelligent agents and multi-agent systems
- Learning and adaptation
- Evolutionary computing
- Neural networks
- Collective intelligence
- Methodologies and applications of intelligent systems

PART B: Your Unit in more detail

Unit Improvements

Feedback provided by previous students through the Student Survey has resulted in improvements that have been made to this unit. Recent improvements include:

- Updates of Project Assignment to include contemporary projects on Deep Learning

Unit Teaching Staff

Name	Role	Room	Email / Teams	Consultation Times
Nguyen, Manh Toan	Unit Convenor in Hanoi		toannguyen@swin.edu.au	By email appointment
Minh Hoang	Unit Convenor in HCM		manh@swin.edu.au	By email appointment
Quang Chiem	Unit Convenor in HCM		qchiem@swin.edu.au	By email appointment
Son Phong Nguyen	Unit Convenor in Danang		sonnguyen@swin.edu.au	By email appointment

Learning and Teaching Structure

Category	Activity	Total Hours	Hours per Week	Teaching Period Weeks
In person /Live Online	Lectures	24 hours	2 hours	Weeks 1 to 12
In person	Tutorials	24 hours	2 hour	Weeks 1 to 12
Online	Self-paced online learning activities	102 hours	8.5 hours	Weeks 1 to 12

The lectures will be in person (on Hawthorn campus, Room BA302) in Weeks 1, 5, 6 and 12. The lectures in Weeks 2, 3, 4, 7, 8, 9, 10 and 11 will be Live Online on Collaborate Ultra (Canvas).

All tutorial classes will be in person (on Hawthorn campus). Please check your timetable for the time + venue of your classes. **Please make sure to attend these classes because your tutor will be checking your progress in your assignments and failure to show the progress will receive significant penalties.**

Week by Week Schedule

Week	Week Beginning	Teaching and Learning Activity	Student Task or Assessment
1	May 11	Topic: Overview. Definition and scope of IS (AI and IA). Philosophical aspects of AI and IA. Tutorial: Getting to know your tutor and teammates; Setting up your computing environment. Java and JADE	Reading unit references. Forming the project team.
2	May 18	Topic: Problem-Solving Agents: Search and Constraint Satisfaction Problem. Tutorial: Choco (Constraint Solver) and JADE.	Reading unit references. Forming the project team and selecting project.
3	May 25	Topic: Multi-agent systems. Agent interactions (encounters, games). Agent communication. Reaching agreements (negotiations). Tutorial: JADE	Reading unit references. Finalising project selection and team. Project assignment work.
4	June 01	Topic: Introduction to machine learning (ML) Tutorial: Introduction to Python & Machine Learning with Python practicals.	Project assignment work. Reading unit references. Continuous Oral Defence.
5	June 08	Topic: Machine learning (ML) algorithms (DT, RF, KNN, PCA) Tutorial: Machine Learning with Python practicals & Project assignment.	Project assignment work. Reading unit references. Continuous Oral Defence.
6	Jun 15	Topic: Artificial neural networks and Deep Learning (DL) Tutorial: ML/DL with Python practicals & Project assignment.	Project assignment work. Reading unit references.
7	June 22	Topic: Computational intelligence. Genetic algorithm and evolutionary computing. Tutorial: ML/DL with Python practicals & Project assignment.	Project assignment progress review. Reading unit references. Continuous Oral Defence.

8	June 29	Topic: Computational intelligence. Fuzzy systems. PSO/ACO. Tutorial: EC/GA with PyGad & Project assignment	Project assignment work. Reading unit references. Continuous Oral Defence.
Summer Holiday – No classes from 06 – 12 July, inclusively.			
9	July 13	Topic: Advanced topic 1: Reinforcement learning. Tutorial: Project assignment.	Project assignment work. Reading unit references.
10	July 20	Topic: Advanced topic 2: Knowledge Representation and Reasoning (KRR) & Expert Systems (ES) or Distributed Constraint Optimization Problems (DCOP). Tutorial: Project assignment.	Project assignment work. Reading unit references.
11	July 27	Topic: Advanced topic 3: Natural Language Processing (NLP). Tutorial: Project assignment.	Project assignment work. Reading unit references. Continuous Oral Defence.
12	Aug 03	Topic: Future directions of AI and IA – selected topics and examples. Review of the subject. Sample examination. Tutorial: Project assignment.	Project Assignment Report, Video, Working Prototype and Presentation Continuous Oral Defence.

Assessment

a) Assessment overview

Tasks and Details	Individual or Group	Weighting	Mapped Unit Learning Outcomes	Mapped Graduate Attributes	Assessment Due Date
1. Project Assessment	Group	50%	1, 2, 3	GA2, GA3, GA5, GA6	Continuous until End of week 12
2. Continuous Oral Defence	Individual	50%	1, 2, 3	GA1, GA5	Continuous until End of week 12/13

To pass this unit, you must:

- achieve an overall mark for the unit of 50% or more.

Assessment Requirements	Details
b) Use of generative AI (genAI) in this unit	The valid use of genAI in this unit is as follows: ● Assessment Task 1: genAI may be used for brainstorming, creating structures, and generating ideas for improving work. Any use of genAI must be acknowledged, with prompts and outputs included in an appendix. ● Assessment Task 2: The assessment should be completed entirely without AI assistance.
c) Hurdle requirements	
d) Final assessment period	If the unit you are enrolled in has a final assessment (including invigilated exams), you will be expected to be available for the entire final assessment period including any Special Exam period. Assessment Task 2 (Continuous Oral Defence) will be conducted face-to-face and may require you to attend a session after Week 12.
e) Submission requirements	Assignments and other assessments are generally submitted online through the Canvas assessment submission system which integrates with the Turnitin . Please ensure you keep a copy of all assessments that are submitted. In cases where a hard copy submission is required an Assessment Cover Sheet must be submitted with your assignment. The standard Assessment Cover Sheet is available from the Submitting work webpage or www.swinburne.edu.au/studentforms/ ● All assessments are to be submitted to Canvas
f) Extensions and late submissions	Late Submissions - Unless an extension has been approved, late submissions will result in a penalty. You will be penalised 10% of your achieved mark for each working day the task is late, up to a maximum of 5 working days. After 5 working days, a zero result will be recorded.

g) Referencing	To avoid breaching academic integrity, you are required to provide references whenever you include information from other sources in your work and acknowledge when you have used Artificial Intelligence (AI) tools (such as ChatGPT). Further details regarding academic integrity are available in Section C of this document. Referencing conventions required for this unit are: Anderson, J. & Poole, M. (2001). <i>Assignment and thesis writing 4th Edn.</i> Brisbane: John Wiley & Sons Helpful information on referencing can be found at http://www.swinburne.edu.au/library/referencing/
h) Groupwork guidelines	A group assignment is the collective responsibility of the entire group, and if one member is temporarily unable to contribute, the group should be able to reallocate responsibilities to keep to schedule. In the event of longer-term illness or other serious problems involving a member of group, it is the responsibility of the other members to notify immediately the Unit Convenor or relevant tutor. Group submissions must be submitted with an Assignment Cover Sheet, signed by all members of the group. All group members must be satisfied that the work has been correctly submitted. Any penalties for late submission will generally apply to all group members, not just the person who submitted.

Required Textbook(s)

The required textbook(s) are available through the Swinburne Library or can be purchased from bookshops.

Course Notes:

Lecture Notes, 2025 (Available from the subject Canvas site during the teaching period)

Text Book:

Negnevitsky, M., 2005. *Artificial intelligence: a guide to intelligent systems*. Pearson education, 3rd edition.

Wooldridge, M. *An Introduction to Multi-Agent Systems*. John Wiley & Sons, 2002

Russell, S. & Norvig, P. *Artificial Intelligence: A Modern Intelligence*. Prentice Hall, 3rd edition, 2009.

Recommended Reading Materials

Swinburne Library has a large collection of resources. Listed below are some references that will provide valuable supplementary information to this unit. It is also recommended that you explore other sources to broaden your understanding.

- B. Coppin, "*Artificial Intelligence Illuminated*" Jones and Bartlett Publishers, 2004
- Poole David L. and Mackworth Alan K. 2023. *Artificial Intelligence: Foundations of Computational Agents*. 3rd Edition, Cambridge University Press, USA.

PART C: FURTHER INFORMATION



For further information on any of these topics, refer to Swinburne's Student webpage <http://www.swinburne.edu.au/student/>

Student behaviour and wellbeing

All students are expected to: act with integrity, honesty and fairness; be inclusive, ethical and respectful of others; and appropriately use University resources, information, equipment and facilities. All students are expected to contribute to creating a work and study environment that is safe and free from bullying, violence, discrimination, sexual harassment, vilification and other forms of unacceptable behaviour.

The [Student Charter](#) describes what students can reasonably expect from Swinburne in order to enjoy a quality learning experience. The Charter also sets out what is expected of students with regards to your studies and the way you conduct yourself towards other people and property.

You are expected to familiarise yourself with University regulations and policies and are obliged to abide by these, including the [Student Academic Misconduct Regulations](#), [Student General Misconduct Regulations](#) and the [People, Culture and Integrity Policy](#). Any student found to be in breach of these may be subject to disciplinary processes.

Examples of expected behaviours are:

- conducting yourself in teaching areas in a manner that is professional and not disruptive to others
- following specific safety procedures in Swinburne laboratories, such as wearing appropriate footwear and safety equipment, not acting in a manner which is dangerous or disruptive (e.g. playing computer games), and not bringing in food or drink
- following emergency and evacuation procedures and following instructions given by staff/wardens in an emergency response.

Canvas

You should regularly log on to the Swinburne learning management system, Canvas. You can access Canvas via the [Student login](#) webpage or <https://swinburne.instructure.com/> Canvas is updated regularly with important unit information and communications.

Communication

All communication will be via your Swinburne email address. If you access your email through a provider other than Swinburne, then it is your responsibility to ensure that your Swinburne email is redirected to your private email address.

Academic Integrity

Academic integrity is about taking responsibility for your learning and submitting work that is honestly your own. It means acknowledging the ideas, contributions and work of others; referencing your sources and acknowledging the use of generative artificial intelligence;

contributing fairly to group work; and completing tasks, tests and exams without cheating. Artificial intelligence tools should only be used where approved by the Unit Convenor.

Swinburne University uses the Turnitin system, which helps to identify inadequate citations, poor paraphrasing and unoriginal work in assignments that are submitted via Canvas. Your Unit Convenor will provide further details.

Plagiarism, collusion, contract cheating, unauthorised file sharing, falsification, fabrication, manipulation or misrepresentation of information, reuse of previous work and non-compliance with instructions in an invigilated or non-invigilated assessment item are all breaches of academic integrity and treated as academic misconduct. Examples of breaches of academic integrity include, but are not limited to:

- submitting work as your own for assessment that has been fully or partially completed by a third party, either paid or unpaid
- using output from artificial intelligence tools (e.g. ChatGPT) in whole or part without acknowledgement and/or without the approval of the Unit Convenor
- using another person's work or ideas as though it is your own work, without appropriate attribution
- working closely with another student or group of students (either past or current), to submit for assessment, some or all of the other student or students' work as your own work
- sharing without permission of the Unit Convenor, Swinburne resources or other material related to assessment to an entity or document repository site
- creating, intentionally modifying or inventing information that is intended to be submitted as part of an assessment item
- using the whole or part of a computer program written by another person as your own without appropriate acknowledgement
- poorly paraphrasing somebody else's work
- using a musical composition or audio, visual, graphic and photographic work created by another person without acknowledgment
- enabling others to cheat, including letting another student copy your work or by giving access to a draft or completed assignment
- letting someone or something else impersonate you, or you impersonate someone else in an invigilated or non-invigilated assessment item
- accessing, obtaining and/or providing to others unauthorised materials relating to an invigilated or non-invigilated assessment item.

The penalties for academic misconduct can be severe, ranging from a zero grade for an assessment task through to exclusion from Swinburne. For further details, see

<https://www.swinburne.edu.au/student-login/academic-integrity/>

Student support

Swinburne offers a range of services and resources to help you complete your studies successfully. Your Unit Convenor or studentHQ can provide information about the study support and other services available for Swinburne students. For further information, see the [Current students](#) web page.

Special consideration

If your studies have been adversely affected due to serious and unavoidable circumstances outside of your control (e.g. severe illness or unavoidable obligation), you may be able to apply for special consideration (SPC).

Applications for Special Consideration are submitted via the SPC online tool normally no later than 5.00pm on the third working day after the submission/sitting date for the relevant assessment component. See

<https://www.swinburne.edu.au/life-at-swinburne/student-support-services/special-consideration-assistance/>

Note: Submitting fraudulent (fake or altered) medical certificates is considered misconduct and can lead to serious penalties from Swinburne. In addition, your doctor may report fraudulent medical certificates as a prosecutable offence under the Victorian Crimes Act.

AccessAbility Services

If you are a student with a disability, medical or mental health condition or you have significant carer responsibilities, you may require reasonable adjustments to fully access and participate in education. Swinburne's AccessAbility Services can develop an Education Access Plan (EAP) that includes the services and reasonable adjustments that you need.

It is recommended that you register with AccessAbility Services when you first commence your course but you can contact the service at any time during your studies to find out about reasonable adjustments. Contact [Accessibility Services](#) to discuss further.

Review of marks

An independent marker reviews all fail grades for major assessment tasks. In addition, a review of assessment is undertaken if your final result is between 45 and 49 or within 2 marks of any grade threshold.

You can ask the Unit Convenor to check the result for an assessment item or your final result. Your request must be made in writing within 10 working days of receiving the result. The Unit Convenor can discuss the marking criteria with you and check the aggregate marks of assessment components to identify if an error has been made. This is known as local resolution. If you are dissatisfied with the outcome of the local resolution, you can lodge a formal complaint.

Feedback, complaints and suggestions

In the first instance, discuss any issues with your Unit Convenor. If your concerns are not resolved or you would prefer not to deal with your Unit Convenor, then you can complete a feedback form. See <https://www.swinburne.edu.au/corporate/feedback/>

Advocacy

If you require assistance with any academic issues, University statutes, regulations, policies and procedures, you are advised to seek advice from an Independent Advocacy Officer at the Swinburne Student Association. Talking to an Advocacy Officer is free, independent and confidential. For more information and booking an appointment, please see <https://www.swinburne.edu.au/current-students/student-services-support/advocacy/>